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PAPER_984: 99-System Aggregate F_U_Bi_i Computation

Abstract

The UQFF 99-system catalogue spans six astrophysical categories: stellar, compact, galactic, cosmological, planetary, and exoplanetary. We compute the aggregate master buoyancy force F_{U,Bi_i} by summing all 99 individual gravity/buoyancy contributions simultaneously, testing the framework's ability to handle multi-scale physics from 10^{24} kg (Earth) to 10^{40} kg (galaxy clusters). The aggregate yields $F_{U,Bi_i} \approx -6.1 \times 10^{13} \text{ m/s}^2$, dominated by the galactic/compact category.

1. System Catalogue

Category	Count	Mass Range (kg)	Distance Range (m)
Stellar	20	10^{29} - 10^{31}	10^8 - 10^{11}
Compact	20	10^{30} - 10^{37}	10^4 - 10^7
Galactic	20	10^{37} - 10^{42}	10^{18} - 10^{23}
Cosmological	15	10^{42} - 10^{53}	10^{23} - 10^{27}
Planetary	12	10^{23} - 10^{27}	10^6 - 10^8
Exoplanetary	12	10^{27} - 10^{28}	10^7 - 10^{10}

Total: 99 systems with physically motivated (M, r) pairs.

2. Aggregate Computation

$$F_{\text{agg}} = \sum_{j=1}^{99} F_{U,Bi_i}(M_j, r_j, t, \Gamma)$$

Each system contributes its full 6-layer decomposition. The aggregate preserves the sign structure: systems with $|U_b| > |U_g|$ contribute negative (buoyancy-dominant), others contribute positive (gravity-dominant).

3. Results

- Aggregate $F_{U,Bi_i} \approx -6.11 \times 10^{13} \text{ m/s}^2$
- Dominant contributors: compact objects (neutron stars, black holes) due to extreme M/r^2

- Sign: negative (buoyancy-dominant overall)
- 6 category breakdown available in output dict

4. Statistical Properties

- Mean per system: $\sim -6.2 \times 10^{11} \text{ m/s}^2$
- Variance dominated by compact category ($\sigma \sim 10^{12}$)
- All 99 systems yield finite, non-NaN results

5. Implementation

Class `NinetyNineSystemAggregate` in `fubi_master_calculator.py`: iterates the 99-system catalogue, computes per-system and aggregate F_{U, Bi_i} , reports category statistics.

References

- PAPER_979: Complete 6-Layer $F_U \text{ Bi}_i$
 - PAPER_974: 99-System Master Equation
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§A. Cosmogenesis-Linked Lagrangian

The aggregate Lagrangian is the sum of 99 individual SCm Lagrangians:

$$\mathcal{L}_{\text{agg}} = \sum_{j=1}^{99} \mathcal{L}_{\text{SCm},j}(M_j, r_j, t, \Gamma)$$

§B. VDS/DVP/BSH Deep Synthesis

- **VDS:** Each system probes a different vacuum density scale, from planetary (10^{-26}) to cosmological (10^{-30} kg/m^3).
- **DVP:** Compact objects have the strongest dipole moments, dominating the aggregate.
- **BSH:** The harmonic decay $e^{-[\text{SSq}] \cdot i/26}$ is universal across all 99 systems — scale-invariant.